Computer Organization (CSD206): Tutorial 2

1. Differentiate between computer structure and computer function.

2. Describe the four main components of any general purpose computer.

3. What do you understand by stored program concept?

4. Explain the basic instruction cycle with the help of state diagram.

5. Define the following processor registers: MAR, MBR, PC, and IR.

6. On the IAS, what would the machine code instruction look like to load the contents of memory address 2 to the accumulator? How many trips to memory does the CPU need to complete this instruction during the instruction cycle? [Hint: Opcode for loading memory contents to AC is 00000001]

7. The hypothetical machine having 16-bit instruction set (4 bits for opcode and 12 bits for address) supports following opcodes:
   - ✓ 0001 = Load AC from memory;
   - ✓ 0010 = Store AC to memory;
   - ✓ 0011 = Load AC from I/O;
   - ✓ 0100 = Multiply to AC from memory;
   - ✓ 0101 = Add to AC from memory;
   - ✓ 0110 = Add to AC from I/O;
   - ✓ 0111 = Store AC to I/O;
   - ✓ 1000 = Multiply to AC from I/O;

The contents of memory and CPU registers in hexadecimal. Consider the initial value of PC to be 200.
Show the contents of processor registers PC, MAR, MBR, I/O AR, I/O BR, AC, IR during the execution of the following program segment:

i. Load AC from memory location 341
ii. Add contents of memory location 340
iii. Store AC to memory location 340
iv. Load AC from memory location 341
v. Multiply contents of device 7
vi. Store AC to memory location 341
vii. Load AC from memory location 340
viii. Multiply contents of memory location 341
ix. Store AC to device 2

Assume that the initial value retrieved from device 7 is 3, memory location 340 contains a value of 2 and memory location 341 contains a value of 1.
Assume address of I/O device 7 as 007 and I/O device 2 as 002.
Memory Stack drawn on Board.